# FortSite Fresh-Start Master Architecture & Delivery Roadmap

**Version:** Fresh-start baseline 3.0
**Planning date:** 26 October 2026
**Product:** FortSite — construction management and ERP for building developers/contractors in Nepal and India
**Status:** Proposed implementation baseline; not a claim that any phase is built, certified, or compliant.

> **Confirmed starting state:** FortSite starts from ZERO: a fresh empty Git repository and a fresh empty Supabase project. There is no existing application code, schema, migration history, authentication setup, role system, component library or completed phase to audit or preserve. Treat this V3 document as the canonical implementation plan. Phase 01 creates the product and architecture documentation only; Phase 02 creates the frontend foundation; Phase 04 establishes the Supabase development/ deployment workflow; later phases create the schema and features in dependency order. Do not import or assume any prior FortSite implementation or legacy database design. Do not reset the empty Supabase project; build it deliberately through versioned migrations.

## 1. Product decision

FortSite is a practical, mobile-friendly system for companies developing and building apartments, villas, small townships, restaurants, hotels and warehouses. The core product must help teams control work, materials, people, procurement and project finances through one connected data model. The product is not meant to be a showcase of specialist enterprise technology.

The implementation starts at Phase 01, with no phase assumed complete. Phases are dependency-ordered delivery increments. A phase is not “done” because a screen or table exists; it is done when its security, workflow, failure and acceptance tests pass.

### In scope for the core product

- Super Admin onboarding and subscription control; company tenancy and permission boundaries.
- Company Admin, Projects Manager, Site Supervisor, Accountant & Billing, and Inventory Manager workspaces.

- Projects/sites/building areas; tasks, schedule, BOQ/estimates, budgets, contracts/variations, DPR and measured progress.
- Workforce, attendance, labour cost tracking, subcontractors, equipment, materials, store inventory and procurement.
- Expenses, native double-entry accounting, accounts payable/receivable, payments/reconciliation, project cost control and reports.
- Basic building quality/safety/commissioning and handover/defects workflows, document storage, import/export and release readiness.
- Supplier Marketplace as a separately entitled add-on with listings, in-app messaging, managed RFQs, structured quote submission/comparison and in-app/mobile push notifications. Supplier relationships and quote workflows remain permissioned and auditable; marketplace is not a payment processor.

### Explicitly not part of this initial roadmap

GIS/chainage surveying, Monte Carlo risk modelling, AI-first workflows, carbon/ESG accounting, roads/hydropower/industrial EPC/quarry/RMC production, face recognition, IoT/telematics, full statutory payroll, full facilities management, real-estate buyer CRM/installment sales, marketplace payments/commissions, automatic supplier order placement, automatic PO creation from quotes, and automatic stock movements from marketplace activity. Automated takeoffs and BIM/IFC clash detection are in scope but must be engineer-reviewed and traceable; they are not a substitute for signed engineering judgement. Government connector activation is in scope only as the final Phase 67 after the rest of the platform passes its release gate. Reconsider any additional specialist capability only after real customer evidence, a separate business decision and a dedicated scope/acceptance plan.

## 2. Technical stack and boundaries

| Layer | Standard | Rules |
|---|---|---|
| Web frontend | React 19, Vite, TypeScript, React Router | Strict types; feature-oriented code; route protection plus server/database authorization; no client secrets. |
| Companion mobile app | React Native + Expo + TypeScript, phased separately | Native Android/iOS app for field tasks, messages, RFQs, notifications and quick field records. Reuse domain types/API contracts where practical; web UI libraries such as shadcn/21st.dev do not automatically run in React Native. |
| Styling/components | Tailwind CSS, shadcn/ui, 21st.dev MCP/plugin in Cursor | Local shadcn components are source-controlled. Use 21st.dev for discovery/install of reviewed UI code, then adapt to FortSite tokens, test it and own the final component. |

| Layer | Standard | Rules |
|---|---|---|
| Backend | Supabase | PostgreSQL, Supabase Auth, Storage, Edge Functions and Realtime only where needed. Supabase is the application data/backend source of truth; no separate custom server by default. |
| Mobile push delivery | Supabase notification/ outbox records + Edge Functions + Expo Push Service (or approved provider selected in Phase 60) | APNs/FCM are ultimately involved for iOS/Android. Push is best-effort delivery, not the authoritative notification ledger; store delivery attempts/results, retry safely and provide an in-app inbox. Keep provider tokens/secrets server-side. |
| Data access/security | Supabase JS API + PostgreSQL RLS + reviewed SQL functions/ Edge Functions | Every exposed tenant table gets RLS/grants reviewed; privileged operations are checked server-side. Never put a service-role key in frontend code. |
| Database workflow | Supabase CLI migrations, seed data and version control | Apply tested migrations in order; do not make undocumented production schema edits. Keep development/staging and production credentials separate. |
| Pooling | Supabase connection pooling/PgBouncer where the access path requires it | Browser Supabase JS uses the Supabase API; do not connect a browser directly to PostgreSQL or ship DB credentials. Use suitable pooled connections for direct/server-side clients. |
| Tests | Typecheck/lint, unit/ component, database/ RLS, integration and end-to-end workflow tests | Test both allowed and forbidden actions. A frontend-only test is not proof of tenant isolation. |

Official technical reference links are listed in Section 12. Current versions and integration steps must be rechecked at implementation time.

# 3. Construction design tokens and UI rules

| Token | Value / rule | Intended use |
|---|---|---|
| Construction Amber | `#F59E0B` | Primary action and restrained emphasis; use dark/Graphite text on amber for contrast. |

| Token | Value / rule | Intended use |
|---|---|---|
| Graphite | `#1F2937` | Main text, dark navigation surfaces and high-contrast controls. |
| Blueprint Blue | `#2563EB` | Links, focus ring and information/action states. |
| Canvas | `#F8FAFC` | Application page background. |
| Surface | `#FFFFFF` | Cards, dialogs and tables. |
| Steel Border | `#E2E8F0` | Dividers, borders and table lines. |
| Semantic states | Accessible success/ warning/danger/info tokens, checked against contrast | State meaning must never depend on color alone; combine with text/icon. |

- **Buttons:** pill-shaped/full-round; consistent heights; clear primary/ secondary/outline/ghost/destructive variants; loading, disabled and keyboard focus states. Avoid white text on amber if contrast is insufficient.
- **Cards:** 12–16px radius; light border; restrained shadow; avoid turning every row into a card.
- **Forms:** short, grouped by task, inline validation, useful defaults, autosaved drafts only where safe, clear submit/approve/post differences.
- **Tables:** search/filter/sort, sticky headers when useful, readable density, responsive fallback, totals linked to detail.
- **Field UX:** quick Today at Site, crew muster, DPR, material request, stock receipt/issue, issue/snags and small expense. Show pending upload/sync distinctly from server-confirmed changes.
- **21st.dev:** search with specific constraints; inspect code/ dependencies/license/accessibility; install once; normalize to tokens; run tests. It is a code-acquisition/design tool, not the source of truth for auth, data or business rules.
- **Progressive disclosure:** only show active, built and relevant modules. Specialist features are not exposed as empty nav links.

# 4. Panel and role division

FortSite uses a shared frontend codebase but separate guarded route zones. Route guards improve navigation; RLS and server-side checks enforce actual data access. The existence of a role or menu item never grants authority on its own.

## Platform zone — Super Admin

**Platform Owner (highest authority):** create/onboard companies; create/invite first Company Admin; choose plan/trial; manage subscription status/dates and record platform subscription payments/ adjustments; activate/suspend/reactivate tenants; manage platform plans/settings, support and audit. Platform Owner is not automatically allowed to browse company operational data as part of routine platform administration. Any future break-glass support access must be explicit, time-limited, reasoned and logged.

Optional restricted Platform Support access can be introduced only when needed; it must not be equivalent to Platform Owner and must not edit subscription plans or access tenant content by default.

## Company zone — five primary roles

| Role | Main responsibilities | Must not automatically do |
|---|---|---|
| Company Admin | Company profile/settings, invites, team assignment, projects overview, company-level configuration and operational visibility. | Create Platform Owners, bypass subscription eligibility, or silently rewrite posted financial/ stock history. |
| Projects Manager | Assigned project planning, BOQ/estimates, budgets, contracts/variations, task/ schedule, progress, inspections, subcontractor scope and project reporting. | Change platform subscription, manage company-wide access by default, or post accounting/stock adjustments without authority. |
| Site Supervisor | Assigned site/area tasks, DPR, progress evidence, attendance/muster, issues/snags, inspections, material requests and expense drafts. | See other projects by default, approve own expense, approve own stock adjustment or access the full ledger. |
| Accountant & Billing | Approved financial documents, project costs, journal/accounting, supplier invoices, client bills, receipts/payments, reconciliation and finance reports. | Alter verified site quantities/attendance without workflow, or access unrelated company tenants. |
| Inventory Manager | Material catalogue, stores, stock receipts/issues/ transfers/counts, requisitions, purchase execution and receiving. | Post unrestricted adjustments, change accounting periods or manage platform subscriptions. |

## Identity, membership, role and scope must stay distinct

1. **Identity:** the Supabase Auth user. 2. **Membership:** the company the user belongs to. 3. **Role:** which actions the user may perform. 4. **Scope:** which project/site/records those actions apply to. 5. **Assignment:** the operational work the user is responsible for. A task assignment must never grant access by itself.

Users may have more than one company role only where it is operationally justified; permissions are the union of explicitly granted actions within the valid company/project/site scope, except for separation-of-duty rules. Keep the first release to named roles and limited overrides, not a large arbitrary role builder.

## Supplier Marketplace zone — separate add-on

Supplier users manage only their own company, listings, messages and quotes addressed to them. Authorized buyer-company users browse published listings, message suppliers in FortSite, create managed

RFQs, collect quotes and compare them within their company. Supplier-specific quotes are never exposed to competing suppliers. The marketplace is not a payment processor. No RFQ or quote automatically creates a PO, payable or stock movement; a permitted user must explicitly approve and create the downstream procurement records.

## 5. Data and security contracts

- Every private company-owned record must be company-scoped and RLS-protected. Add project/site scope where the domain needs it; validate cross-table tenant consistency using database constraints/functions rather than trusting frontend IDs.
- RLS applies to exposed tables; object access in Supabase Storage is permissioned as well. Test read, create, update, delete and relationship paths for allowed and forbidden actors.
- Never expose service-role keys, direct DB passwords, webhook secrets or payment-provider secrets in the Vite bundle. Use Supabase secrets and Edge Functions/server-side logic for privileged operations.
- Platform authority and company authority are separate. Company Admin cannot grant platform access. A route guard is not a security boundary.
- Cost-bearing records must retain transaction currency and link to the source, company, project/site where applicable, cost code where applicable, creator and approval/posting state.
- Nepal defaults: NPR and Asia/Kathmandu. India defaults: INR and Asia/Kolkata. Do not convert FX or combine unlike currencies into a single total. Changing a default never rewrites prior transactions.
- Financial/inventory postings must be atomic and idempotent; posted records are corrected by reversal/adjustment, not silent mutation. Preserve source-document-to-ledger links.
- Stock is driven by a ledger of posted movements, not a user-editable on-hand field. Accounting journals must balance and posting paths must be tested against duplicate retries.
- Store timestamps in UTC and display in company timezone. Historical records retain the timezone/currency/tax context required to explain them.
- Realtime subscriptions are used selectively for notification/work-queue freshness, not as a substitute for authorization or reliable reads.
- Subscription status/entitlements are checked against trusted server/database state. JWT/UI state alone must not decide whether a company is active.
- Marketplace messages, RFQs, quote documents, quote prices, supplier responses, push tokens and notification payloads are private records. Apply participant/tenant/supplier-side RLS and storage rules; a supplier can access only RFQs addressed to that supplier and their own quotes.
- Mobile push notifications contain minimal sensitive data; deep links re-authorize through the app. The notification inbox/outbox is the durable source of truth, not push delivery status.
- Automated quantity/rebar outputs must carry source drawing/model revision, input assumptions, units, formula/rules version, reviewer

and approval status. They remain draft/unverified until an authorized engineer/estimator approves them.

- IFC model files and clash results are permissioned project data. Do not use Supabase Edge Functions for heavy geometric processing by default; test browser/Web Worker/WASM processing on representative models before choosing the execution path.
- Country/registered jurisdiction is captured and validated during company onboarding. Nepal and India use separate tax/compliance profiles and provider adapters. Government credentials are encrypted in server-only storage, never exposed to browser logs, client state or ordinary user-facing records.

# 6. Global definition of done

Every phase must meet all relevant checks below before being marked complete: 1. Defined user workflow and acceptance criteria are met with realistic data. 2. Typecheck, lint, unit/component and production build pass; applicable end-to-end tests pass. 3. Database changes are in reviewed migrations; RLS and Storage policies are included and tested. 4. Negative permissions are tested, including cross-company and cross-project attempts. 5. Loading, empty, failure, duplicate-submit, expired-session and retry states are handled. 6. Audit history exists for approvals, privileged changes and financial/stock-sensitive actions. 7. No financial/stock side effect can be posted twice through retry; history is not silently overwritten. 8. UI matches the tokens, has keyboard-visible focus, appropriate contrast and mobile usability. 9. Documentation explains how to run, test, migrate and roll back the changed part. 10. Phase status is recorded as NOT STARTED / IN PROGRESS / BLOCKED / VERIFIED COMPLETE with linked evidence; do not infer completion from a menu item.

# 7. Full phase register and implementation plan

**Total: 67 phases (01–67).** All start as NOT STARTED in a fresh implementation. Dependencies should be honored; later business features must not be started ahead of the foundations they rely on. The national government billing connectors are deliberately the final numbered phase.

## Phase index

| Phase | Name | Dependencies |
|---|---|---|
| 01 | Product charter, workflow requirements and architecture blueprint | None; this is the project starting point. |
| 02 | Repository bootstrap and engineering standards | Phase 01. |
| 03 | Construction design system and 21st.dev component governance | Phase 02. |

| Phase | Name | Dependencies |
|---|---|---|
| 04 | Supabase environments, migrations and deployment pipeline | Phases 01–02. |
| 05 | Authentication, profiles and session lifecycle | Phases 02 and 04. |
| 06 | Company tenancy, memberships and tenant-safe data foundations | Phases 04–05. |
| 07 | Secure platform-owner identity and privileged operations boundary | Phases 04–06. |
| 08 | Super Admin panel shell and platform overview | Phases 02–03 and 07. |
| 09 | Company onboarding and lifecycle management | Phases 06–08. |
| 10 | Subscription plans, entitlements and manual payment records | Phases 07–09. |
| 11 | Platform audit, settings and support operations | Phases 07–10. |
| 12 | Company roles, project/site scope and permission matrix | Phases 05–07 and 09. |
| 13 | Company first-run setup and company settings | Phases 06, 09 and 12. |
| 14 | Company workspace shell and role-based home screens | Phases 03, 05, 06, 12 and 13. |
| 15 | Approvals, notifications and company activity history | Phases 06, 11–14. |
| 16 | Shared master data, units, categories and cost codes | Phases 06, 12–13. |
| 17 | Secure document, drawing, photo and attachment storage | Phases 04, 06, 12 and 14. |
| 18 | Spreadsheet import, data export and migration utilities | Phases 12–17. |
| 19 | Project portfolio, projects and sites | Phases 06, 12–16. |
| 20 | Building, block, floor, unit and work-area hierarchy | Phase 19. |
| 21 | Client, consultant and vendor contact register with simple proposals | Phases 13, 16–19. |

| Phase | Name | Dependencies |
|---|---|---|
| 22 | Work breakdown structure, work packages and project cost codes | Phases 16, 19–20. |
| 23 | Task assignment and accountability | Phases 14–16 and 19–22. |
| 24 | Rate library for materials, labour, equipment and work items | Phases 16, 19 and 22. |
| 25 | BOQ, quantities, estimates and revision history | Phases 16, 18, 22 and 24. |
| 26 | Project budget and approved baseline | Phases 15, 22, 24–25. |
| 27 | Contracts, work orders, variations and change orders | Phases 15, 19, 22, 25–26. |
| 28 | Practical schedule and weekly look-ahead | Phases 22–23 and 25–27. |
| 29 | Mobile-first field workspace and bounded offline drafts | Phases 03–05, 14–17, 19 and 23. |
| 30 | Daily progress report and site diary | Phases 15, 17, 19, 23, 28–29. |
| 31 | Measured quantities and progress against BOQ | Phases 15, 17, 22, 25, 28–30. |
| 32 | Issues, delays, snag lists and corrective actions | Phases 17, 19–20, 23 and 29–31. |
| 33 | Quality inspections, testing and commissioning checklists | Phases 17, 19–20, 29–32. |
| 34 | Practical safety and permit register | Phases 17, 19, 23 and 29. |
| 35 | Employee, worker and crew register | Phases 12–16 and 19. |
| 36 | Attendance, muster, shifts and overtime records | Phases 15, 19, 29 and 35. |
| 37 | Labour costs, wages, advances and allowances | Phases 16, 19, 35–36 and 48. |
| 38 | Subcontractor agreements and work orders | Phases 15–17, 19, 22 and 27. |
| 39 | Subcontractor measurements, claims and settlements | Phases 15, 27, 38 and 48–50. |
| 40 | Equipment and vehicle register | Phases 16, 19 and 21. |
| 41 | Equipment utilization, fuel, maintenance and cost | Phases 15, 19, 40 and 48. |

| Phase | Name | Dependencies |
|---|---|---|
| 42 | Materials catalogue and specifications | Phases 16–19, 21 and 24. |
| 43 | Site stores, store locations and inventory ledger | Phases 06, 15–17, 19 and 42. |
| 44 | Stock issue, consumption, return, count and waste | Phases 15, 19, 29–30 and 42–43. |
| 45 | Material requisitions and procurement quote capture | Phases 15–16, 19, 21 and 42–44. |
| 46 | Purchase orders, commitments and receiving | Phases 15–16, 19, 42–45 and 48. |
| 47 | Site expenses, petty cash, advances and reimbursements | Phases 15, 17, 19 and 48–51. |
| 48 | Native double-entry accounting and posting engine | Phases 04, 06, 10–16 and the first operational financial workflows. |
| 49 | Supplier invoices and accounts payable | Phases 15, 21, 46–48. |
| 50 | Client billing, progress bills and accounts receivable | Phases 15, 25–27, 31 and 48. |
| 51 | Cash, bank, payment allocation and reconciliation | Phases 15, 48–50. |
| 52 | Integrated budget, commitments, actual cost and simple forecast | Phases 22, 25–27, 37–51. |
| 53 | Nepal/India tax, invoice and document rules | Phases 13 and 48–51. |
| 54 | Role dashboards, operational reports and safe exports | Relevant source phases 15–53. |
| 55 | Project closeout, handover, warranties and defect liability | Phases 17, 20, 32–34, 38–39 and 54. |
| 56 | Supplier Marketplace add-on: supplier profiles, product catalogue and moderation | Phases 05–12, 14, 16–17, 21 and 42–46. |
| 57 | Marketplace in-app messaging and conversation controls | Phases 06, 12, 17 and 56. |
| 58 | Managed RFQ creation, supplier invitations and deadlines | Phases 15–17, 21, 42, 45, 56–57. |
| 59 | Structured supplier quote submission, comparison and selection | Phases 25–27, 45–52, 56 and 58. |

| Phase | Name | Dependencies |
|---|---|---|
| 60 | Notification centre, delivery outbox and mobile push service | Phases 05–06, 15 and 57–59. |
| 61 | React Native/Expo mobile companion app | Phases 02–05, 12, 14, 17, 23, 29–30, 35–36 and 56–60. |
| 62 | Drawing-based automated quantity takeoffs | Phases 17, 20, 22, 25, 31 and 42. |
| 63 | Automated rebar quantities and bar-bending schedules | Phases 16–17, 20, 22, 24–26, 31 and 62. |
| 64 | IFC model import, model viewer and revision management | Phases 17, 20, 22, 25 and 32. |
| 65 | BIM/IFC clash detection and coordination workflow | Phases 17, 32–33 and 64. |
| 66 | Complete end-to-end validation, restore drills, pilot and release readiness | Phases 01–65. Government production connectors remain disabled; use test adapters/fixtures. |
| 67 | FINAL: Nepal IRD/CBMS and India IRP government billing integrations | Phases 09, 13, 48–54 and 66; live activation requires approved authority/provider path and applicable credentials. |

## Detailed phase specifications

### Phase 01 — Product charter, workflow requirements and architecture blueprint

**Dependencies:** None; this is the project starting point.
**Scope:** Create one agreed definition of what FortSite does for Nepal/India building developers and contractors. Document the real workflows for apartments, villas, townships, hotels, restaurants and warehouses; identify actors, inputs, decisions, outputs, exceptions and acceptance tests. Establish the entity map, route map, security assumptions, non-functional requirements and change-control rules before code is written.
**Explicit exclusions:** No UI implementation, operational database tables, or broad feature invention. Do not carry old phase numbers or old architecture forward merely because they exist.
**Exit gate:** Product owner signs off on scope; every core workflow has a measurable acceptance scenario; every feature is marked core, separate add-on, deferred/unapproved, or explicitly out of scope.

### Phase 02 — Repository bootstrap and engineering standards

**Dependencies:** Phase 01.
**Scope:** Create the React 19 + Vite + TypeScript application foundation, strict TypeScript rules, React Router, Tailwind CSS, shadcn/

ui, linting, formatting, unit-test setup and a production build. Establish a feature-oriented code layout, typed route definitions, reusable error/loading/empty states, environment validation and a readable README. Use compatible current stable packages at implementation time and commit the lockfile.
**Explicit exclusions:** No business modules, mock operational workflows that look production-ready, giant one-file components, unchecked `any`, direct database credentials in the browser, or unreviewed dependency pile.
**Exit gate:** Fresh clone installs reproducibly; development server, lint, typecheck, tests and production build pass; unknown routes and runtime errors have deliberate behavior.

## Phase 03 — Construction design system and 21st.dev component governance

**Dependencies:** Phase 02.
**Scope:** Create central theme tokens and a small design system before building product screens. Construction palette: Amber #F59E0B, Graphite #1F2937, Blueprint Blue #2563EB, canvas #F8FAFC, white surfaces, steel borders #E2E8F0, and accessible semantic states. Buttons use pill/full-round shape, with graphite text on amber primary buttons, visible focus states, disabled/loading states and consistent sizing. Cards use 12–16px radius and restrained shadow. Define typography, spacing, tables, badges, dialogs, forms, alerts and mobile breakpoints. Use 21st.dev MCP to find/install reviewed components, then adapt them to FortSite tokens and local accessibility standards.
**Explicit exclusions:** Do not let each screen use its own palette, install multiple competing component systems, use 21st.dev code without reviewing its dependency list/license/accessibility, or treat 21st.dev as a runtime backend. Prefer existing local shadcn components before installing duplicates.
**Exit gate:** A documented token file and component gallery exist; buttons, forms, tables, dialogs and navigation look consistent; keyboard focus and contrast are checked; no business logic is hidden inside copied presentation components.

## Phase 04 — Supabase environments, migrations and deployment pipeline

**Dependencies:** Phases 01–02.
**Scope:** Set up Supabase local development and separate non-production/production environments as practical. Version all schema changes in migration files; use seed data for development; document how to reset local data, review migrations, deploy Edge Functions and roll back safely. Validate Vite environment variables at startup. Configure the Supabase JS client with only the public project URL and publishable/anon key. Use the API from the browser rather than exposing direct Postgres credentials; use pooling/connection strategies only for suitable server-side or direct database clients.
**Explicit exclusions:** No schema changes directly on production as an undocumented habit, no service-role key in Vite variables, no production write access for casual AI tooling, and no confusion between the browser’s API client and direct PostgreSQL connections.
**Exit gate:** A new developer can reproduce the schema and app from version control; secrets are absent from built assets; migration process and environment ownership are documented.

## Phase 05 — Authentication, profiles and session lifecycle

**Dependencies:** Phases 02 and 04.
**Scope:** Implement Supabase Auth login, logout, password reset, invitation acceptance, protected routes and session restoration. Keep the auth identity separate from the user's profile and tenant membership. Add clear loading/error states and safe handling for expired sessions. Choose the supported sign-in methods based on business operations; do not create public company access by default.
**Explicit exclusions:** No passwords stored by FortSite, no frontend-only role authority, and no assumptions that a signed-in user automatically belongs to any company or is a platform administrator.
**Exit gate:** Sign-in/recovery/invitation/session-expiry flows pass tests; an authenticated user without valid authorization sees no protected data.

## Phase 06 — Company tenancy, memberships and tenant-safe data foundations

**Dependencies:** Phases 04–05.
**Scope:** Create the core company/tenant schema, membership records, invitations, company identity/status fields, unique tenant codes and helper functions for trusted access checks. Define company-owned records to carry company_id and use tenant-consistent foreign-key patterns where relevant. Enable RLS on every exposed tenant table and test company A against company B. Make company eligibility a server/database-enforced rule, not a hidden frontend assumption.
**Explicit exclusions:** No cross-company shared table containing private data without an explicit design, no company_id supplied by the browser trusted without validation, no broad `USING (true)` policies on private records, and no role inferred only from a person being assigned to a project.
**Exit gate:** Tenant isolation tests cover select/insert/update/delete and relationship edge cases; policies are reviewed against actual query paths; the schema supports a user with memberships in more than one company if allowed by product rules.

## Phase 07 — Secure platform-owner identity and privileged operations boundary

**Dependencies:** Phases 04–06.
**Scope:** Create the platform-admin/owner authorization boundary. Bootstrap the first Platform Owner through a controlled, documented process, not public self-registration. Enforce checks in PostgreSQL functions and/or Edge Functions for privileged operations; keep secrets server-side. The Platform Owner is the highest authority for onboarding companies and managing platform subscriptions, but platform authority must not silently turn ordinary requests into company-data access.
**Explicit exclusions:** No role set by user-editable profile fields, no platform-admin creation by a normal Company Admin, no `service_role` in the browser, and no invisible company impersonation. Any future break-glass access must require a reason, time limit and immutable audit trail.

**Exit gate:** Unauthorized users cannot call platform operations by changing route, request payload or client state; the initial owner setup and recovery procedure are documented and tested.

## Phase 08 — Super Admin panel shell and platform overview

**Dependencies:** Phases 02–03 and 07.
**Scope:** Build a separate platform route zone and navigation, e.g. `/platform/*`, with its own authorization guard, layout, accessible tables, filters, confirmation dialogs and responsive behavior. Provide overview cards that drill down to real data: company counts by status, active/trial/overdue subscriptions and recent platform events. Keep this an operations console rather than a marketing dashboard.
**Explicit exclusions:** No company operational workspace in the platform shell, no fake metrics, no decorative charts without source records, and no normal route link that exposes data beyond the user's platform permission.
**Exit gate:** Only Platform Owner (and separately granted restricted platform operators if later enabled) can access this zone; metrics reconcile to underlying records; loading/empty/error states are present.

## Phase 09 — Company onboarding and lifecycle management

**Dependencies:** Phases 06–08.
**Scope:** Allow Platform Owner to create a company, capture and validate its legal registration country (Nepal or India), choose default currency, enter business profile/contact details, create or invite its first Company Admin, select a subscription/trial, and activate, pause, suspend or archive the company under defined rules. Create a country-specific compliance-profile record and route the Company Admin into the appropriate Nepal tax/CBMS setup area or India GST/e-invoice/IRP setup area; at this stage capture profile/eligibility metadata only, not live government credentials or submissions. Generate company identifiers safely. Show onboarding state and a clear reason when a company is not eligible for normal use. The actual government/provider credential linking and live endpoint activation are held for final Phase 67.
**Explicit exclusions:** No open self-service tenant creation unless separately approved; no company deletion that cascades away accounting/audit history; no silent lifecycle transitions.
**Exit gate:** Platform Owner can onboard a test company end-to-end; the invited admin lands in the correct company; status transitions are validated and audited.

## Phase 10 — Subscription plans, entitlements and manual payment records

**Dependencies:** Phases 07–09.
**Scope:** Create plan definitions, subscription terms, start/end dates, trial/grace/active/overdue/suspended/cancelled statuses and a server-side entitlement check. Let Platform Owner change a company's plan and record payments/adjustments with amount, currency, date, reference and notes. Support monthly/annual/lifetime terms only with explicit rules; an annual plan must have an end date. Begin with auditable manual payment recording; integrate payment gateways

only after a separate business decision.
**Explicit exclusions:** No payment card storage, no presumed automatic gateway integration, no frontend-only entitlement enforcement, no currency conversion, and no silent deletion/rewriting of historical subscription payments.
**Exit gate:** Expired/suspended eligibility is enforced on protected company operations; the appropriate billing/renewal explanation remains available; plan changes are audited and history is preserved.

## Phase 11 — Platform audit, settings and support operations

**Dependencies:** Phases 07–10.
**Scope:** Add immutable platform audit events for onboarding, status/plan changes, subscription adjustments, privileged settings and access changes. Provide carefully scoped platform settings and a simple support-ticket/notes workflow if operationally required. Record actor, action, target, time, result and reason for privileged changes. Separate platform logs from company activity history.
**Explicit exclusions:** No routine editing/deleting audit events, no global secret/config editor accessible to non-owners, and no unrestricted access to company documents through support tickets.
**Exit gate:** Sensitive platform actions are auditable and searchable; permissions are tested; settings have validation and safe defaults.

## Phase 12 — Company roles, project/site scope and permission matrix

**Dependencies:** Phases 05–07 and 09.
**Scope:** Define five primary company roles: Company Admin, Projects Manager, Site Supervisor, Accountant & Billing, and Inventory Manager. Keep platform roles completely separate. Support assignment to one or more roles where justified and scope project/site operational access. Membership identity, assigned responsibility and permission are different concepts. Start with a clear fixed role model and only a small set of explicit permission overrides if real use requires them.
**Explicit exclusions:** No sprawling arbitrary role builder in the first release; no automatic permission grant from assignment; no role enum changes without migration and full authorization tests; no ability for Company Admin to grant platform access.
**Exit gate:** A readable resource/action/scope matrix exists; direct URL/API attempts cannot bypass it; cross-project and cross-site negatives are tested.

## Phase 13 — Company first-run setup and company settings

**Dependencies:** Phases 06, 09 and 12.
**Scope:** Give the Company Admin a guided setup for business profile, country-locked compliance profile, company timezone, financial year, default currency, document numbering, logo, business contacts, team invitations and simple approval defaults. Nepal defaults to NPR and Asia/Kathmandu; India defaults to INR and Asia/Kolkata. Route the user into the country-specific billing/compliance settings and show connection status (not configured/pending/active) as appropriate. Permit NPR/INR only where the company workflow allows it and store currency on financial records. Legal jurisdiction is not a casual display

setting; changing it after regulated invoices exist requires a controlled migration/legal-entity workflow.
**Explicit exclusions:** No foreign-exchange conversion; changing a default must not rewrite historical transaction currencies; no giant settings form containing features the company does not use.
**Exit gate:** A Company Admin can configure a new tenant without support; company-specific settings are tenant-isolated and changes are audited.

## Phase 14 — Company workspace shell and role-based home screens

**Dependencies:** Phases 03, 05, 06, 12 and 13.
**Scope:** Build the company route zone, such as `/app/:companyId/*`, with tenant context derived from verified membership. Create the navigation and role-aware landing screens for Company Admin, Projects Manager, Site Supervisor, Accountant & Billing and Inventory Manager. Show work queues and tasks for that role instead of exposing every future module at once. Include accessible global search affordance once relevant data exists and consistent breadcrumbs/context.
**Explicit exclusions:** No role-based access implemented only by hiding menu items; no empty links to unbuilt modules; no one dashboard forced on all roles.
**Exit gate:** Each core role has a useful first screen, a minimal menu and a tested guard; smaller screens remain usable; inactive modules are not advertised as implemented.

## Phase 15 — Approvals, notifications and company activity history

**Dependencies:** Phases 06, 11–14.
**Scope:** Provide a simple shared approval pattern for expenses, requisitions/purchase orders, contract variations, measured work and bills as needed. Support submit, approve, reject/return with reason, timestamped history and actionable in-app notifications. Make actions idempotent so retrying does not double-post money or stock. Keep common flows configurable through modest thresholds and named approvers.
**Explicit exclusions:** No universal no-code workflow engine, no uncontrolled notification spam, and no ability to approve one’s own record when separation of duties is required.
**Exit gate:** Approval transitions, notifications, unread state and audit history are tested; repeated commands do not duplicate downstream effects.

## Phase 16 — Shared master data, units, categories and cost codes

**Dependencies:** Phases 06, 12–13.
**Scope:** Implement reusable, searchable masters for units of measure, material categories, labour categories, equipment categories, cost codes, tax labels, payment methods and document sequences. Support standard defaults plus company-specific values. Prevent duplicates where practical and preserve old references when a category is archived.
**Explicit exclusions:** No general-purpose custom database/form

designer or dozens of mandatory master-data screens before the first project can be created.
**Exit gate:** Core modules use the same IDs and definitions; units/ categories validate inputs; archived values remain visible on historical records.

## Phase 17 — Secure document, drawing, photo and attachment storage

**Dependencies:** Phases 04, 06, 12 and 14.
**Scope:** Use Supabase Storage with private buckets and object policies linked to company/project/site permissions. Store metadata, uploader, file type, size, capture time and optional document category/version. Generate short-lived access URLs when appropriate. Support resumable/retryable uploads where the chosen client library permits and clear upload state for field users.
**Explicit exclusions:** No sensitive object in a public bucket by default; no guessing storage paths for access control; no unpermissioned model/drawing access. IFC processing and automated takeoffs are delivered only in Phases 62–65.
**Exit gate:** Unauthorized tenant/user cannot list, read, replace or delete another company's files; attachment links honor permissions; upload failure/retry is clear.

## Phase 18 — Spreadsheet import, data export and migration utilities

**Dependencies:** Phases 12–17.
**Scope:** Provide validated imports for the practical starting data: clients/vendors, workers, materials, project metadata, BOQ rows, budgets and opening stock where supported. Preview rows, map columns, show validation errors and allow safe retry without duplicate records. Export approved lists and reports to CSV/XLSX/PDF where implemented; record import batch and outcomes.
**Explicit exclusions:** No opaque one-click import that silently corrupts ledgers; no unbounded export of records the user cannot read; no claiming every competitor's data format is supported.
**Exit gate:** Representative spreadsheet files can be imported with row-level feedback and reconciliation counts; duplicate/retry behavior is safe; export totals match source records.

## Phase 19 — Project portfolio, projects and sites

**Dependencies:** Phases 06, 12–16.
**Scope:** Create projects with code, client, type, dates, location, manager, status and commercial context. Represent sites as operational locations under projects where required; allow one project to have multiple sites and one company to have multiple projects. Include archive/close lifecycle, filters, status history and drill-downs.
**Explicit exclusions:** No assumption every company has only one project/site; no hard delete of projects with financial/operational history; no map/GIS dependency.
**Exit gate:** Projects and sites are isolated by company and relation integrity prevents attaching another company's site to a project; lifecycle is covered by tests.

## Phase 20 — Building, block, floor, unit and work-area hierarchy

**Dependencies:** Phase 19.
**Scope:** Support building-oriented project spaces: blocks/buildings, floors, apartments/units, rooms/zones and common areas. Let a simple villa project skip unnecessary hierarchy, while hotel/apartment/warehouse projects use it when helpful. Allow work packages and inspections to reference the right building area without making every field mandatory.
**Explicit exclusions:** No space-planning CAD or forced room register for small projects that do not need one. Specialist IFC model viewing is handled later in Phase 64 and must not be mandatory for simple projects.
**Exit gate:** A project may operate with or without the hierarchy; room/unit lists can be imported and filtered; records never cross project/company boundaries.

## Phase 21 — Client, consultant and vendor contact register with simple proposals

**Dependencies:** Phases 13, 16–19.
**Scope:** Maintain company contacts for clients, architects/consultants, contractors and suppliers, with contact details and project links. Support a lightweight opportunity/proposal record where useful to the construction firm: scope summary, estimate reference, submitted date, status and attached proposal. Keep supplier contacts usable in procurement without requiring the Marketplace add-on.
**Explicit exclusions:** No real-estate buyer CRM, buyer payment plans, marketing automation or full sales-management suite in the core roadmap.
**Exit gate:** Contacts are reusable across projects; proposal state and attachments are permissioned; Marketplace contacts remain distinct from internal procurement records unless a user explicitly creates/links a vendor record.

## Phase 22 — Work breakdown structure, work packages and project cost codes

**Dependencies:** Phases 16, 19–20.
**Scope:** Build a practical breakdown of project scope into stages/trades/work packages and assign cost codes that connect estimates, BOQ, tasks, material, labour, subcontractors and financial postings. Include common templates for building stages while allowing project-level edits. Show roll-up totals with traceability to source lines.
**Explicit exclusions:** No complex Primavera-grade critical-path dependency matrix or mandatory work breakdown setup for every tiny job.
**Exit gate:** A work package can be traced from planned scope to execution and costs; roll-ups reconcile to detail; archived codes preserve historical references.

## Phase 23 — Task assignment and accountability

**Dependencies:** Phases 14–16 and 19–22.
**Scope:** Create tasks with owner/crew, project/site/area, planned dates, priority, status, due date, notes, checklist and links to supporting

records. Make assignment and overdue tasks visible in the relevant role's Today/Work Queue view. Support comments or activity notes only where they serve a real execution need.
**Explicit exclusions:** No project-management mega-suite with unnecessary chat channels, automation canvases or dozens of task fields.
**Exit gate:** A task can be created, assigned, completed, reopened with reason where permitted, and tracked by project/site; users see only authorized tasks.

## Phase 24 — Rate library for materials, labour, equipment and work items

**Dependencies:** Phases 16, 19 and 22.
**Scope:** Provide reusable rate items with unit, currency, effective date/source and optional region/vendor. Rates feed estimates, BOQ and budgets but remain snapshots in approved historical documents. Allow company defaults and project overrides with traceable edits.
**Explicit exclusions:** No external live-price scraper, dynamic AI pricing or silent updates to already-approved estimates.
**Exit gate:** Rates are unit-consistent; historical quotes/estimates retain their original rates; currency is explicit and never silently converted.

## Phase 25 — BOQ, quantities, estimates and revision history

**Dependencies:** Phases 16, 18, 22 and 24.
**Scope:** Create a building-project BOQ with item code/description, unit, quantity, rate, amount, cost code, specification and revisions. Support estimate preparation, grouped work sections and comparison of revisions. Allow practical spreadsheet import/export. Distinguish draft estimate from approved baseline.
**Explicit exclusions:** No unreviewed quantities that silently alter an approved BOQ. Automated drawing takeoffs and rebar schedules are implemented later in Phases 62–63; initial BOQ entry/import remains fully usable without them.
**Exit gate:** BOQ quantity x rate totals validate; revision compare works; approved quantities/rates are not silently changed; source and approval metadata are present.

## Phase 26 — Project budget and approved baseline

**Dependencies:** Phases 15, 22, 24–25.
**Scope:** Create an approved project budget tied to BOQ/work packages/cost codes, with revisions and contingency/allowance lines where needed. Record baseline version, approver and effective date. Give Projects Manager and Accountant a shared reference for cost control.
**Explicit exclusions:** No Monte Carlo forecasts or predictive AI. Keep contingency simple and explicit.
**Exit gate:** Budget line totals reconcile; changing an approved baseline creates a new version/approval rather than overwriting history.

## Phase 27 — Contracts, work orders, variations and change orders

**Dependencies:** Phases 15, 19, 22, 25–26.
**Scope:** Track owner/client contracts, subcontract/work orders, agreed scope/value, start/end dates, retention/advance where applicable and linked change orders. A change request records reason, value/time impact, supporting evidence, approval status and effect on the approved project baseline only after approval.
**Explicit exclusions:** No full legal contract drafting suite or electronic-signature provider integration unless separately approved.
**Exit gate:** Contract/variation changes have revision and approval history; approved value changes flow consistently to budget/billing; draft changes never silently affect posted finance.

## Phase 28 — Practical schedule and weekly look-ahead

**Dependencies:** Phases 22–23 and 25–27.
**Scope:** Provide a simple task schedule with planned start/finish, dependencies for common sequencing, responsible crew and weekly look-ahead. Show late tasks and baseline versus actual dates. Allow manual adjustments and readable timeline/list views.
**Explicit exclusions:** No 4D/5D model, Monte Carlo schedule risk, resource-optimization solver or advanced Primavera clone.
**Exit gate:** A project manager can maintain the schedule and weekly look-ahead; actual dates/history are preserved; schedule status links to execution tasks.

## Phase 29 — Mobile-first field workspace and bounded offline drafts

**Dependencies:** Phases 03–05, 14–17, 19 and 23.
**Scope:** Optimize common site workflows for Android/mobile web: Today at Site, assigned tasks, DPR, attendance/muster, material request, issue/snags, inspection and small expense draft. Cache only the minimum safe data. Allow draft capture and a clear pending-upload/sync queue when connectivity is poor; the server confirms approval/posting.
**Explicit exclusions:** No promise of complete offline ERP, offline posting of accounting/inventory, silent conflict resolution or background sync that can duplicate transactions.
**Exit gate:** A user can save drafts during interrupted connectivity, restart and resume safely, see sync failures/conflicts, and never confuse a draft with a server-confirmed record.

## Phase 30 — Daily progress report and site diary

**Dependencies:** Phases 15, 17, 19, 23, 28–29.
**Scope:** Capture date/site, weather as optional, workforce count, work completed, quantities, materials used/received, equipment used, delays, instructions, incidents and photo evidence. Provide a short repeat-use form with sensible defaults and auto-filled project/site/user/date. Allow submit and review workflow.
**Explicit exclusions:** No mandatory long narrative for every day and

no AI-generated DPR requirement.
**Exit gate:** A normal DPR can be drafted quickly; submitted reports are traceable; corrections keep history; evidence is stored privately.

## Phase 31 — Measured quantities and progress against BOQ

**Dependencies:** Phases 15, 17, 22, 25, 28–30.
**Scope:** Record measured work quantities against BOQ/work packages, date, location, measurement notes, evidence, preparer and verifier. Roll up completed versus planned quantities and estimate progress using clearly defined quantities/weights rather than arbitrary percentage inputs alone. Keep accepted measures separate from draft measurements.
**Explicit exclusions:** No advanced survey chainage, drone measurement or quantity inference from ordinary site photos. Drawing-based assisted/automated takeoffs are a separate workflow in Phase 62.
**Exit gate:** Progress totals are reproducible from accepted measurements; duplicate entries and over-measurement are flagged; project/site/BOQ links validate.

## Phase 32 — Issues, delays, snag lists and corrective actions

**Dependencies:** Phases 17, 19–20, 23 and 29–31.
**Scope:** Manage site issues with category, severity, responsible person, due date, status, location, photos and resolution evidence. Support delay reason and impact notes; link defects/snags to room/unit, task, inspection or contractor. Provide filters by open/overdue/area/responsible party.
**Explicit exclusions:** No enterprise claims-management engine. Model-based clash coordination is delivered later as a separate workflow in Phases 64–65 and is not required to use ordinary site issue/snags.
**Exit gate:** Issues can be logged on mobile, assigned, updated, resolved and reopened with history; overdue queues work; access is scope-controlled.

## Phase 33 — Quality inspections, testing and commissioning checklists

**Dependencies:** Phases 17, 19–20, 29–32.
**Scope:** Provide configurable templates for common building quality inspections: concrete/rebar/formwork checks where used, waterproofing, masonry/plaster, finishes, electrical/plumbing/MEP tests and pre-handover checks. Capture pass/fail/not-applicable, evidence, inspector, date, corrective issue and sign-off.
**Explicit exclusions:** No specialist lab-information system or automated building-control integration; do not make every project use every checklist.
**Exit gate:** Templates are reusable; failed checks create or link corrective actions; approved inspection results are immutable except through a documented correction.

## Phase 34 — Practical safety and permit register

**Dependencies:** Phases 17, 19, 23 and 29.
**Scope:** Track site inductions or toolbox talks when used, incidents/near misses, corrective actions, selected risk/permit checklists, responsible person and evidence. Keep it usable for building projects and local practice, and make configurable forms optional. High-risk work approvals can be simple checklists, not a new platform.
**Explicit exclusions:** No massive EHS enterprise suite, automated regulatory reporting promises or permit workflows that cannot be supported locally.
**Exit gate:** Incident records are protected, follow-up actions are assigned, and safety records are searchable/exportable with an audit trail.

## Phase 35 — Employee, worker and crew register

**Dependencies:** Phases 12–16 and 19.
**Scope:** Maintain workers/employees and crews with name, contact/reference ID, trade/skill, employment/vendor relationship, status and documents as appropriate. Minimize sensitive personal data and restrict it by role. Support contractor-provided crews without assuming all workers are direct employees.
**Explicit exclusions:** No biometric/face-recognition system, broad HR suite or mandatory collection of sensitive identity information not needed for the workflow.
**Exit gate:** Worker records are company-scoped; crew lists can be bulk-imported; inactive workers remain available on historical records.

## Phase 36 — Attendance, muster, shifts and overtime records

**Dependencies:** Phases 15, 19, 29 and 35.
**Scope:** Record daily attendance by site and crew using bulk muster; support present/absent/half-day, shift and overtime fields where used. Enable correction requests/approval and link the record to labour cost calculations. Optional location/photo evidence must be governed and never be the only proof of attendance.
**Explicit exclusions:** No face recognition or geofencing by default; no payroll-compliance claims based on attendance alone.
**Exit gate:** Representative crews can be marked in bulk; corrected attendance retains history; downstream cost calculations use approved records only.

## Phase 37 — Labour costs, wages, advances and allowances

**Dependencies:** Phases 16, 19, 35–36 and 48.
**Scope:** Track agreed wage/rate basis, approved days/hours, overtime/allowances when applicable, advances, deductions and payment records, with a clear balance per worker/crew and project cost allocation. Allow export to a payroll provider or accountant if required, but keep FortSite's core scope as construction labour cost/settlement tracking.
**Explicit exclusions:** No statutory payroll engine, income-tax withholding computation, social-security contribution engine or promise to replace a country-compliant payroll product.

**Exit gate:** Worker/crew liabilities and project labour-cost totals reconcile to approved attendance and payments; edits to paid history create an adjustment entry.

## Phase 38 — Subcontractor agreements and work orders

**Dependencies:** Phases 15–17, 19, 22 and 27.
**Scope:** Manage subcontractor profiles, scope/work package, agreement amount, rate basis, start/end date, retention/advance terms, documents and authorized contacts. Connect each subcontract to one or more projects and cost codes. Keep commercial terms visible to authorized company roles only.
**Explicit exclusions:** No end-to-end subcontractor portal in the initial core unless a separately approved requirement is documented.
**Exit gate:** Subcontract scope/value is linked to budget and measurement; access is restricted; contract changes require history and approval.

## Phase 39 — Subcontractor measurements, claims and settlements

**Dependencies:** Phases 15, 27, 38 and 48–50.
**Scope:** Capture measured work/claim lines, quantity, rate, deductions/retention/advance recovery, supporting evidence, review/approval and payment status. Approved claims feed accounts payable and project costs through a single linked posting path. Keep draft claim, approved amount and paid amount distinct.
**Explicit exclusions:** No duplicate payable created by approving and then retrying; no silent changes after posting.
**Exit gate:** Claim totals calculate correctly; approval/posting is atomic or idempotent; reversal/correction trail is tested.

## Phase 40 — Equipment and vehicle register

**Dependencies:** Phases 16, 19 and 21.
**Scope:** Track owned/rented equipment and vehicles: identifier, type, owner/provider, rate/cost basis, condition, documents, project/site allocation and status. Start with an ordinary register, not an asset-accounting or fleet telematics system.
**Explicit exclusions:** No GPS telematics, IoT tracking or complex depreciation engine in the initial build.
**Exit gate:** Equipment can be assigned to projects/sites; key dates and condition are searchable; history is preserved.

## Phase 41 — Equipment utilization, fuel, maintenance and cost

**Dependencies:** Phases 15, 19, 40 and 48.
**Scope:** Record equipment usage hours/days, operator if needed, fuel, routine service, repairs, breakdowns and project allocation. Surface upcoming service and downtime with simple lists. Link costs to project cost codes when appropriate.
**Explicit exclusions:** No predictive maintenance AI, live fleet tracking

or machine telemetry.
**Exit gate:** Usage and cost reports reconcile to entries; maintenance reminders are not presented as verified work until completed.

## Phase 42 — Materials catalogue and specifications

**Dependencies:** Phases 16–19, 21 and 24.
**Scope:** Create reusable material items with code, name, category, unit, specification/grade/brand, common pack size, preferred supplier references and optional images/documents. Support project-specific alternatives and clear unit conversions where mathematically valid. Keep an internal company catalogue independent of Marketplace listings.
**Explicit exclusions:** No live supplier price scraper or assumption that marketplace product data is approved procurement data.
**Exit gate:** Material item and UOM definitions are consistent across BOQ, requisitions, receipts, issues and costs; archival does not damage history.

## Phase 43 — Site stores, store locations and inventory ledger

**Dependencies:** Phases 06, 15–17, 19 and 42.
**Scope:** Represent stores/warehouses and location-level stock. Use a movement ledger as the authoritative record of receipts, issues, transfers, returns, approved adjustments and opening balances. On-hand quantities are derived from posted movements or maintained transactionally with reconciliation controls. Model partial receipts and multiple sites/stores.
**Explicit exclusions:** No editable on-hand number as the sole source of truth; no automatically moving stock from quotations or Marketplace clicks.
**Exit gate:** Ledger balances reconcile to the movement history; company/store scope is secured; concurrent transactions do not silently create negative/inconsistent stock.

## Phase 44 — Stock issue, consumption, return, count and waste

**Dependencies:** Phases 15, 19, 29–30 and 42–43.
**Scope:** Support issuing materials to a task/site/work package/crew, recording consumption or return, cycle counts, damaged/wasted stock, transfer and discrepancy approval. Differentiate ordered, received, accepted, issued, returned and consumed quantities. Capture simple reason codes and cost allocation.
**Explicit exclusions:** No hidden stock change when a DPR is submitted; no unrestricted inventory adjustment by every user.
**Exit gate:** Each movement has actor, date, quantity, unit, destination/source and source document; stock is not duplicated when a command is retried.

## Phase 45 — Material requisitions and procurement quote capture

**Dependencies:** Phases 15–16, 19, 21 and 42–44.
**Scope:** Let site staff request materials with required-by date, project/site/store, cost code, requested quantity and reason. Reviewers

approve/return; procurement can record supplier quotations manually, compare price/unit/lead time and select a vendor. Procurement can record supplier quotations received manually for ordinary procurement. When the separately entitled Marketplace sourcing workflow is enabled, Phases 57–59 provide managed in-app messages, RFQs and quote collection; both paths remain available.
**Explicit exclusions:** No automatic PO creation from a catalogue listing, or stock movement from a requisition/quote. Managed Marketplace messaging/RFQs/quotes are added in Phases 57–59 and do not replace internal approval or PO creation.
**Exit gate:** Approved requisitions trace to quote comparison and eventual PO; quantities/units and approvals validate; external Marketplace contact is clearly separate.

## Phase 46 — Purchase orders, commitments and receiving

**Dependencies:** Phases 15–16, 19, 42–45 and 48.
**Scope:** Create approved purchase orders with line items, supplier, prices, currency, taxes/charges where applicable, delivery details and status. Receive items partially or fully through a goods-received record; record accepted/rejected/short quantities and link accepted stock to the inventory ledger. Track outstanding quantity and commitments until matched/closed.
**Explicit exclusions:** No direct stock update from an unapproved PO or unverified supplier contact; no duplicate receipt from retry.
**Exit gate:** PO, receipt, inventory movement, payable and committed-cost records are linked; partial receiving and discrepancy tests pass.

## Phase 47 — Site expenses, petty cash, advances and reimbursements

**Dependencies:** Phases 15, 17, 19 and 48–51.
**Scope:** Make small everyday expense capture fast, including food/tea/snacks, local transport, unloading, tools, fuel and emergency purchases. Use date, amount, currency, category, project/site/cost code, paid-by, payment source, note and optional receipt photo. Support advances, settlement and approval, with a short mobile form and draft save.
**Explicit exclusions:** No ten-step entry for small spends; no automatically approved expense because the amount is low unless the company explicitly configures a bounded rule.
**Exit gate:** A common repeat-use expense is quick to draft; submitted/approved/paid states are distinct; one approved expense produces at most one posting.

## Phase 48 — Native double-entry accounting and posting engine

**Dependencies:** Phases 04, 06, 10–16 and the first operational financial workflows.
**Scope:** Implement a reviewed chart of accounts, journal headers/lines, financial periods, debit-credit balance validation, source-document links, posting status, reversal entries and period locking. Integrate operational source documents through atomic/idempotent posting operations. Keep accounting as FortSite’s target system of record, with accounting-specialist review before financial production use.

**Explicit exclusions:** No unbalanced journals, no direct edit/delete of posted journal lines, no silent posting from frontend multi-step inserts, and no assumption the software is formally compliant without review.
**Exit gate:** Every posted journal balances; source and journal links are traceable; duplicate commands do not duplicate postings; reversal and period-lock tests pass.

## Phase 49 — Supplier invoices and accounts payable

**Dependencies:** Phases 15, 21, 46–48.
**Scope:** Record supplier invoices against PO/receipts or approved non-PO expenses where policy allows. Track invoice lines, due date, taxes/charges, advances, credits, approved amount, payment status and project/cost-code allocations. Support partial matching and duplicate invoice warnings.
**Explicit exclusions:** No unreviewed automatic 3-way matching assumptions; no double-paying an invoice; no supplier-marketplace transaction record inferred from a listing.
**Exit gate:** Invoice totals and outstanding balance reconcile; invoice/receipt/PO relationships are validated; posting to accounts payable is idempotent.

## Phase 50 — Client billing, progress bills and accounts receivable

**Dependencies:** Phases 15, 25–27, 31 and 48.
**Scope:** Prepare client bills from contract milestones, measured work or approved BOQ quantities, with retention/advance recovery/variations when applicable. Track submitted, approved, due, part-paid, paid, disputed and cancelled/reversed states. Link bills, receipts and accounting postings; retain previous bill revisions.
**Explicit exclusions:** No automatically sending tax invoices or treating draft measurement as billable/approved; no silent rewrite of issued invoices.
**Exit gate:** Billing math and outstanding balances are tested; approved quantities are sourced and traceable; corrections use revision, credit/reversal or adjustment procedures.

## Phase 51 — Cash, bank, payment allocation and reconciliation

**Dependencies:** Phases 15, 48–50.
**Scope:** Record cash/bank accounts, receipts, payments, allocations against invoices/advances and bank-statement reconciliation. Support partial and combined payments with references and proof. Keep transaction currency explicit; surface unmatched or unreconciled items.
**Explicit exclusions:** No access to bank credentials, no automatic bank feeds unless separately approved, and no mixing NPR/INR in a single misleading total.
**Exit gate:** Payments allocate once; bank reconciliation differences are explainable; cash/bank balances trace to journals and source transactions.

## Phase 52 — Integrated budget, commitments, actual cost and simple forecast

**Dependencies:** Phases 22, 25–27, 37–51.
**Scope:** Show budget versus approved changes, PO/subcontract commitments, actual posted costs and a transparent estimate-to-complete or cost-to-complete entry. Break down by project, site, work package and cost code. Indicate what is posted versus draft/committed and link every total to detail.
**Explicit exclusions:** No AI prediction or Monte Carlo simulation; no combining unlike currencies or unposted records into a misleading actual-cost number.
**Exit gate:** Budget/commitment/actual totals reconcile to source modules; forecast assumptions are visible; dashboard values drill down to records.

## Phase 53 — Nepal/India tax, invoice and document rules

**Dependencies:** Phases 13 and 48–51.
**Scope:** Add country-configured invoice numbering/fields, tax configuration and export formats only after qualified local accounting/tax review. Store tax amounts and rules with each source transaction; retain historical values when settings change. Document which invoice types and rules the product actually supports for Nepal versus India.
**Explicit exclusions:** No live connection to government billing endpoints in this phase; no claims of IRD/CBMS or Indian GST/IRP compliance without applicable technical approvals/testing; no hard-coded one-country tax assumptions reused in the other; no automatic legal advice. Live Nepal/India government connector work is deliberately deferred to final Phase 67.
**Exit gate:** Country-specific sample scenarios are reviewed by qualified practitioners; invoice calculations and numbering tests pass; unsupported tax cases are clearly blocked or flagged.

## Phase 54 — Role dashboards, operational reports and safe exports

**Dependencies:** Relevant source phases 15–53.
**Scope:** Provide practical prebuilt reports: project status, planned vs actual progress, DPR history, workforce attendance, material stock/usage, open requisitions/POs, equipment use, open issues/inspections, expenses, budget vs commitments vs actual, AP/AR and cash/bank reconciliation. Filters and totals must match source data. Exports obey the same permission scope as on-screen records.
**Explicit exclusions:** No data warehouse, no-code BI builder or decorative KPIs that do not reconcile.
**Exit gate:** Each report has a defined formula/source, access test and export test; totals match detail on representative data.

## Phase 55 — Project closeout, handover, warranties and defect liability

**Dependencies:** Phases 17, 20, 32–34, 38–39 and 54.
**Scope:** Create a practical closeout checklist per building/project, with room/unit snag completion, final inspections, commissioning, as-built/

document handover, warranty records, manuals, keys/asset handover and defect-liability dates/contacts. Track post-handover defects and responsibility within a simple record set.
**Explicit exclusions:** No full property/facilities-management product, tenant service desk or property-owner CRM.
**Exit gate:** A project can progress through closeout with incomplete items visible; handover package and outstanding defects are reportable; final documents retain their permissions and history.

## Phase 56 — Supplier Marketplace add-on: supplier profiles, product catalogue and moderation

**Dependencies:** Phases 05–12, 14, 16–17, 21 and 42–46.
**Scope:** Build a separately entitled marketplace route and supplier panel. Suppliers maintain a public business profile, service areas, contact information and listings including name, category/type, brand/model, grade, weight/dimensions, specification, unit/pack, price, currency, MOQ, availability, last-updated date, product images and technical documents/certifications. Include search, useful filters, listing moderation, verification status, report/stale-listing handling and audit history. Publish only approved listings. Supplier records/listings and company vendor records remain distinct unless a user deliberately links them.
**Explicit exclusions:** No marketplace payment processing, commission ledger, automatic order, automatic PO, payable or stock movement. Messaging and RFQ workflows are separate phases.
**Exit gate:** Supplier users can only manage their own supplier profile/listings; company users see published listings only; moderation is auditable; add-on entitlements are enforced; listing freshness and availability are visible.

## Phase 57 — Marketplace in-app messaging and conversation controls

**Dependencies:** Phases 06, 12, 17 and 56.
**Scope:** Implement direct buyer-supplier conversations and context-linked conversations from a supplier listing or project/RFQ. Store durable message records, sender, participants, timestamp, read state, optional attachments, moderation/report/block controls and retention policy. Use Supabase Realtime only for live updates; message history remains database-backed. Give a user clear unread/read and failed-send/retry states. Keep each supplier conversation private to its participants and authorized company users.
**Explicit exclusions:** No public group chat, cross-supplier shared threads, message content exposed through push payloads, or promise that a read receipt proves a human read the message.
**Exit gate:** Participants see only authorized threads/messages/attachments; retries do not duplicate messages; blocked/reported accounts have defined behavior; thread permissions are tested from both buyer and supplier sides.

## Phase 58 — Managed RFQ creation, supplier invitations and deadlines

**Dependencies:** Phases 15–17, 21, 42, 45 and 56–57.
**Scope:** Allow authorized company users to convert an internal material/procurement need into a managed Request for Quotation

(RFQ). Select products/categories and invited suppliers, specify line items, quantity/unit, delivery location/date, scope/specifications, attachments, response deadline, terms and internal approval requirements. Suppliers receive an RFQ in their own panel, accept/decline or ask permitted clarifying questions, and respond within the RFQ. Track draft, awaiting approval, issued, response received, closed, cancelled and awarded/no-award states. Use stable IDs and timestamped status history.
**Explicit exclusions:** No automatic PO creation; no supplier inclusion without explicit invite/visibility rules; no claim that marketplace pricing is a binding quote unless a supplier submits it as one; no bypassing existing company procurement approval thresholds.
**Exit gate:** Only permitted company roles can issue RFQs; invited suppliers can see only their assigned RFQ and allowed shared documents; dates/status/deadlines work; closed/cancelled RFQs cannot accept silent edits.

## Phase 59 — Structured supplier quote submission, comparison and selection

**Dependencies:** Phases 25–27, 45–52, 56 and 58.
**Scope:** Suppliers submit quote revisions with line-level unit, quantity, rate, taxes/charges as applicable, validity, lead time, delivery terms, exclusions, attachments and notes. Buyers can see each invited supplier's submission, compare normalized line-item values with visible assumptions, record evaluation notes, shortlist/reject/select with approver and reason, and preserve all quote revisions. Do not normalize unlike currencies into a single converted total; comparison must show currency and assumptions. Record selection/award outcome and link it to the internal requisition. A buyer explicitly creates a PO through the normal procurement phase when ready.
**Explicit exclusions:** No auto-award, AI-selected supplier, hidden ranking formula, marketplace escrow/payment or automatic procurement posting.
**Exit gate:** Suppliers cannot access competitors' quotes; buyer comparison totals reconcile to source lines; quote updates retain history; no quote creates a financial/stock ledger movement; the PO handoff is a deliberate authorized action.

## Phase 60 — Notification centre, delivery outbox and mobile push service

**Dependencies:** Phases 05–06, 15 and 57–59.
**Scope:** Extend the notification centre into a durable notification service for message received, RFQ issued/updated/deadline approaching, quote received, RFQ status changed, approval needed and important project work. Create an idempotent event/outbox pattern and preferences by category, company membership and device. Send mobile push via Supabase Edge Functions through an approved push provider (initial recommendation: Expo Push Service with APNs/FCM behind it); record tickets/receipts where the provider supports them, retry transient failures with backoff, deactivate invalid tokens and provide an in-app inbox fallback. Keep notification payloads minimal and never include quote pricing or sensitive message text in lock-screen payloads by default.
**Explicit exclusions:** No requirement that a user enable push, no guarantee of delivery if OS permissions/provider/device disallow it, no

push provider secret in frontend code, and no use of push delivery as the audit source of truth.
**Exit gate:** Authorized user receives the right event only; duplicates are idempotent; user preferences and revoked tokens are respected; provider errors/retries are observable; in-app notifications remain available when push fails.

## Phase 61 — React Native/Expo mobile companion app

**Dependencies:** Phases 02–05, 12, 14, 17, 23, 29–30, 35–36 and 56–60.
**Scope:** Create a separate Android/iOS mobile app using React Native, Expo and TypeScript while retaining the React/Vite web app for administration, financial work and model review. Prioritize high-frequency mobile workflows: authentication/session handling, assigned projects/sites/tasks, DPR capture, photos/attachments, attendance/muster where authorized, material requests, issues/snags, Marketplace messages, RFQ responses/quote upload and push notification deep links. Reuse shared types/API contracts where appropriate, enforce Supabase RLS just like the web app, securely store session material using platform-safe storage and maintain a clear offline/pending/upload-failed state for supported drafts. Configure app identifiers, signing/build workflow and Android/iOS push credentials in documented environments.
**Explicit exclusions:** No assumption that shadcn/ui/21st.dev web components work natively; no duplicate business logic trusted only on-device; no promise that every heavy finance/BIM action belongs on mobile; no unbounded offline replication of sensitive records.
**Exit gate:** Android and iOS build from documented setup; login/session expiry, deep links, attachment upload/retry, role/scope checks and push registration pass; notification taps route to a record that is re-authorized server-side.

## Phase 62 — Drawing-based automated quantity takeoffs

**Dependencies:** Phases 17, 20, 22, 25, 31 and 42.
**Scope:** Add a dedicated drawing workspace for PDF drawings and supported vector/CAD-derived inputs selected after a technical spike. Support sheet/page selection, scale calibration for raster/vector drawings, count/length/area measurements, measurement overlays, assembly/rule-based quantity extraction where geometry and symbols are machine-readable, grouping by drawing/level/zone/BOQ/work package and export of proposed takeoff lines. Keep source drawing revision, page, geometry/measurement, rule version, unit conversion, assumptions and reviewer attached to each result. Label machine-extracted, manually corrected and approved values separately. Allow a user to continue manual BOQ entry/import without this module.
**Explicit exclusions:** No claim that arbitrary scanned drawings are perfectly understood; no silent BOQ/budget edits; no result becomes approved or billable until reviewed; no image recognition on ordinary site photos; no automatic takeoff based only on a marketing-grade model.
**Exit gate:** A defined reference set of common building drawings is

checked against independently measured quantities; discrepancies, unsupported symbols and calibration uncertainty are visible; results trace to source geometry and can be revised without losing history.

## Phase 63 — Automated rebar quantities and bar-bending schedules

**Dependencies:** Phases 16–17, 20, 22, 24–26, 31 and 62.
**Scope:** Calculate reinforcement quantities and draft bar-bending schedules (BBS) from supported structured reinforcement input, approved drawings/schedules and/or supported model data. Record bar mark, diameter, shape, dimensions, count, cut length, hooks/bends, lap/anchorage allowances where specified, unit weight basis, total length and estimated weight; group by element, level/zone, pour or bar diameter. Keep standards/code factors and assumptions explicit and configurable only after qualified engineering review. Preserve drawing/model version, calculation/rule version, estimate vs verified status and engineer/estimator approval. Feed approved quantities to a new takeoff/BOQ revision only through an explicit action.
**Explicit exclusions:** No unsupervised structural design, no inference of missing reinforcement design, no silent substitution of code rules, no claiming the schedule is safe for fabrication before qualified approval, and no automatic purchase order/stock movement.
**Exit gate:** Test cases cover common slabs, beams, columns and footings, bar shapes, laps/hooks, unit conversions and rounding; results are compared to independently prepared schedules; unrecognized/incomplete input blocks final approval.

## Phase 64 — IFC model import, model viewer and revision management

**Dependencies:** Phases 17, 20, 22, 25 and 32.
**Scope:** Add a separately enabled project-model workspace for supported IFC files. Upload/version models to private Supabase Storage; store project/discipline/author/date/revision and file hash; parse/render supported IFC geometry and properties in the browser using a suitable reviewed WebAssembly/Web Worker library after a performance spike. Support model visibility, basic camera/navigation, element/property selection, floor/discipline filters, model revision comparison at a practical level and links from model elements to project areas, tasks, BOQ lines or issues where valid. Preserve the source model unmodified.
**Explicit exclusions:** No full authoring-grade BIM editor, no native Revit replacement, no 4D/5D simulation, no requirement that ordinary projects upload IFC, no reliance on Supabase Edge Functions to process huge geometry payloads unless feasibility is proven.
**Exit gate:** Supported sample IFCs render with validated coordinates/properties; unsupported/corrupt files fail safely; storage access is permissioned; memory/response performance is measured against agreed model-size limits; model revisions are auditable.

## Phase 65 — BIM/IFC clash detection and coordination workflow

**Dependencies:** Phases 17, 32–33 and 64.
**Scope:** Run hard-clash and configured clearance/interference checks across supported IFC models/disciplines using an agreed tolerance/unit

system. Create saved clash runs tied to exact model revisions and settings; group/triage results; create coordination issues with element references, viewpoints/snapshots, location/level, responsible discipline/person, due date, status, discussion, resolution evidence and re-test outcome. Allow filters/exclusions for approved known conditions. Give project teams a basic review/resolve/re-run loop; do not require BIM workflows for non-BIM jobs. Keep expensive geometry processing in the execution environment proven by a benchmark (initially Web Worker/WASM or a specifically approved compute option); only persist model metadata, results, issues and file assets in Supabase.
**Explicit exclusions:** No automatic clash resolution, no guarantee every design conflict is detected, no specialist 4D/5D planning, no engineering approval based solely on clash status, and no automatic change to BOQ/contract/PO.
**Exit gate:** Reference models produce expected positive and negative clash cases; tolerances/units are visible; every result can be reproduced against a model revision; issues follow normal permission/assignment workflow; large-model performance and limits are documented.

## Phase 66 — Complete end-to-end validation, restore drills, pilot and release readiness

**Dependencies:** Phases 01–65. Government production connectors remain disabled; use test adapters/fixtures.
**Scope:** Consolidate security/usability evidence across all modules: tenant/RLS tests, platform boundary tests, Storage policies, Edge Function authorization/secret checks, Marketplace message/RFQ/quote privacy, push token and mobile permissions, quote-to-PO handoff, takeoff/BBS engineer review, IFC model access/clash reproducibility, migration/rollback rehearsal, backups/restore drills, performance tests on representative project/IFC data, accounting reconciliation and error monitoring. Pilot with real building-project users in Nepal and India; migrate an agreed sample; train each role; record task-completion/friction/adoption evidence; resolve critical blockers. Use test/mock government connectors only to verify the tax-adapter contract, invoice payload preparation and failure states.
**Explicit exclusions:** No live official government submission before Phase 67; no claiming final regulatory integration is complete; no release while cross-tenant access, data-loss, unbalanced journals, duplicate financial/stock postings, private quote leakage or unreproducible critical takeoff/clash errors remain open.
**Exit gate:** Product owner signs release-readiness checklist; restore evidence and end-to-end reconciliation pass; pilot feedback is addressed; all critical/high-risk security and data-integrity findings are closed or explicitly blocking; test connector contracts are verified.

## Phase 67 — FINAL: Nepal IRD/CBMS and India IRP government billing integrations

**Dependencies:** Phases 09, 13, 48–54 and 66. This is the final numbered integration/launch phase. Product architecture, invoice/billing workflows, country profiles, test suite, security, pilot and release-readiness must already be complete.
**Scope:** Coordinate with the relevant authority, approved integration route/provider and customer tax administrators. Confirm currently applicable technical documentation, eligibility/mandate, onboarding

requirements, API credentials, sandbox/test endpoints, data fields, signature/QR/IRN requirements where applicable, cancellation/credit/debit note behavior, retries, rate limits, audit evidence, support escalation and production approval. Implement the Nepal-specific IRD/CBMS connector and India-specific GST e-invoice/IRP connector behind a common internal provider contract, with separate adapters, schemas, validation, credentials and observability. During company onboarding, validated legal registration country selects the matching Nepal or India compliance profile and UI; do not let the country be freely switched after regulated invoices exist. The Phase 67 live setup associates the company with the correct authority/provider route and securely links or provisions required government/provider credentials. Store secrets only in server-side encrypted storage; never log them or expose them to browser clients. Test with official sandbox/test credentials first; then run a limited, explicitly approved customer pilot; reconcile FortSite invoices to returned identifiers/status/QR/evidence; deploy production only when written approval/eligibility and a rollback/manual fallback are established.

**Nepal path:** Validate the current IRD CBMS API technical document and the company's eligibility/permission for electronic billing. Treat API approval, taxpayer setup and current CBMS procedures as authority-confirmed prerequisites; do not assume one software approval automatically enables every taxpayer.

**India path:** Confirm the taxpayer's GST/e-invoice applicability and the authorized access path. Do not assume there is one universal provider endpoint for all Indian companies: direct API integration, an IRP/provider's API-integrator route, or another permitted partner route must be selected and authorized for each supported arrangement. Separate taxpayer credentials from FortSite/provider credentials, follow the provider's onboarding and access authorization, and implement any required GSTIN-specific configuration.

**Explicit exclusions:** No scraping of government portals, no bypass of MFA/OTP/authorization, no shared credentials across unrelated taxpayers, no false claim of government certification, no credentials stored in ordinary profile tables/plain text, and no automatic issuance while credentials/authorization/eligibility are uncertain. If official approval is delayed, keep the connector in disabled/pending status and provide the approved manual/export fallback; do not block the entire construction ERP.

**Exit gate:** Current official requirements are documented with dates/contacts; provider/authority access is approved; sandbox test cases pass; security review confirms secret handling; duplicates/retries/cancellations/rejections/timeouts are covered; issued invoices reconcile to the authority's returned status and identifiers; customer-specific credentials are scoped and audited; written go-live authorization and rollback/manual fallback are recorded. No later numbered feature phase follows this integration phase.

# 8. Core workflow acceptance scenarios

- **Tenant isolation:** A user from Company A attempts to read/change Company B projects, files, subscription details, stock, accounting or Marketplace-private records. The request must fail at the database/server boundary, not merely hide the UI.

- **Super Admin onboarding:** Platform Owner creates a company, selects country/defaults and a plan, invites the first Company Admin, records trial or payment terms, activates it and can see the full audit trail. The Company Admin cannot open `/platform/*` or mutate subscription controls.
- **Subscription eligibility:** A suspended/expired tenant cannot continue protected operational mutations, but the user sees a clear status/explanation and any deliberately permitted billing/contact route remains available.
- **Daily site workflow:** Site Supervisor opens assigned site, views tasks, drafts a DPR with quantities and photos, marks attendance, creates a material request and records a small expense. Each record is visible in the correct review queue.
- **Material procurement:** Approved request -> manual supplier quotation comparison -> approved PO -> partial goods receipt -> accepted stock movement -> supplier invoice -> payable. Outstanding quantity, commitment and stock balance reconcile throughout.
- **Inventory traceability:** A storekeeper issues materials to a work package, returns unused stock and records a damaged quantity. On-hand is explainable from movement history; retrying a request never duplicates stock movements.
- **Progress to client bill:** A measured quantity is prepared, checked and accepted; project progress updates from accepted quantities; an approved client bill references only supported billable quantities and produces a traceable receivable.
- **Subcontractor settlement:** An approved subcontractor measurement/claim calculates agreed value/deductions/retention, posts once, becomes payable and tracks partial payment without changing prior approved history.
- **Accounting integrity:** Every posting has equal debits and credits; source document and journal are linked; posted documents cannot be silently edited; reversals retain history; period lock prevents unauthorized back-posting.
- **Field connectivity failure:** A site user loses connectivity while drafting a DPR/expense, closes/reopens the app, retries upload and resolves conflict if necessary. Pending drafts never show as approved or posted.
- **Marketplace messaging/RFQ/quote privacy:** A buyer messages a supplier, issues an RFQ to selected suppliers, receives supplier-specific quote revisions and compares them internally. A supplier cannot see competing suppliers or unrelated company projects. Message/quote retries do not duplicate records; push failure leaves a durable in-app notification; a selected quote never creates a PO/payable/stock movement without an explicit authorized procurement action.
- **Automated takeoff and BBS review:** A drawing is versioned; calibrated/model-derived takeoff outputs link to source locations and assumptions; a qualified reviewer approves/rejects/edits results; only explicitly approved quantities can become a BOQ revision or downstream billable measure. Rebar schedule test cases reconcile to independently prepared reference calculations.
- **IFC clash review:** A clash run records exact IFC model revisions, tolerances and supported units; expected clashes are found and known non-clashes remain absent; results become permissioned

project issues; re-running after a model revision retains historic run results.

- **Country billing routing:** On company onboarding, legal registration jurisdiction selects Nepal or India tax/compliance profile and settings. Production official connectors remain disabled until final Phase 67 approval; a company cannot bypass the country-specific route by editing client state.
- **Closeout:** A project can identify open inspections/snags, outstanding documents, commissioning status, warranty details and defect-liability dates before handover. Unfinished items remain visible and assigned.

# 9. Data-domain boundary map

| Domain | Principal records | Source-of-truth rule |
|---|---|---|
| Platform | Platform owners, companies, plans, subscriptions, subscription payments/adjustments, platform audit, support | Platform Owner manages tenant lifecycle and licence status; company users cannot modify platform records. |
| Identity/access | Auth users, profiles, memberships, invitations, role assignments, project/site scope | Auth user is not automatically a company member; assignment is not permission. |
| Projects/buildings | Projects, sites, blocks, floors, units, rooms/zones, contacts, WBS, tasks, schedule | Operational hierarchy links to a valid company/project and is versioned where baseline matters. |
| Scope/commercial | Rate library, BOQ revisions, estimates, budgets, contracts, variations, measurements | Approved revisions are preserved; changes affect baseline only after approval. |
| Field execution | DPRs, media, progress records, issues, inspections, safety, workforce, attendance | Field evidence is traceable; accepted/verified records are distinct from drafts. |
| Inventory/procurement | Materials, stores, inventory movements, requisitions, quotes, POs, goods receipts, issues/returns | Stock ledger movements are authoritative; PO/receipt/invoice/payment linkages are explicit. |
| Finance | Accounts, periods, journals, supplier invoices, client bills, receipts/payments, reconciliations | Journal lines balance; source records post atomically/idempotently; no silent mutation after posting. |
| Marketplace add-on | Supplier profiles/listings/media, moderation, message threads/messages, RFQs/invitations, quote revisions/comparisons, notification outbox | Add-on records are tenant- and participant-scoped. Suppliers see only their own invitations/quotes; buyer quotes are confidential. No automatic PO/payment/stock effects. |
| Mobile notifications | Notification events, inbox items, device tokens, preferences, delivery attempts/receipts | Supabase records remain authoritative; push is best effort and carries minimal sensitive content. |

| Domain | Principal records | Source-of-truth rule |
|---|---|---|
| Automated construction quantities | Drawing revisions, calibration, measurement geometry, takeoff runs/lines, rebar schedules and review events | Every result traces to source drawing/model and calculation version; only approved results feed formal BOQ/billing. |
| BIM/IFC coordination | IFC model files/versions, element metadata, clash runs/results, coordination issues | Source model is immutable and versioned; results tie to model revisions/tolerances; access follows project permissions. |
| Government billing adapters | Jurisdiction profiles, provider configuration, credential references, submissions, identifiers/status/reconciliation events | Country-specific adapters are separate; secrets are server-only; official connection is enabled only after Phase 67 approval. |

# 10. Practical implementation order

1. **Make the foundation boring and trustworthy:** Phases 01–17. Do not race toward dashboards or finance screens before tenancy, RLS, authorization, migrations, tokens and document access are proven.
2. **Make a project usable:** Phases 18–34. A real building team should be able to set up a project, plan work, record daily progress, measure quantities and close quality/safety issues.
3. **Control people, suppliers, stock and equipment:** Phases 35–47. Prove real receiving/issue/cost workflows before treating inventory as an on-hand number.
4. **Introduce financial system-of-record behavior deliberately:** Phases 48–53. Financial postings need accounting review, idempotency tests and country-specific validation—not just attractive tables.
5. **Finish core reporting and handover:** Phases 54–55.
6. **Deliver optional advanced sourcing and technical tools:** Phases 56–65. Marketplace messaging/RFQ/quotes are a separately entitled add-on; mobile apps handle push and high-frequency field flows; takeoff/rebar and IFC/clash are opt-in specialist modules with engineer-review gates.
7. **Prove readiness before regulatory live connections:** Phase 66 completes the full test, restore, security and pilot gate using test adapters. Phase 67 is deliberately last and covers the country-specific government-authority/provider integration and controlled go-live.

# 11. Change control and scope discipline

- Each phase is implemented and reviewed separately. Do not bundle several phases into one giant AI coding prompt.
- Before writing code for a phase, inspect the relevant files and current migrations, summarize the intended change and identify

security/data risks. Never assume an older conversation or plan proves current state.

- If a prerequisite is missing, stop the dependent implementation and report the blocker; do not create a shortcut that weakens tenant isolation or data integrity.
- A request to add a feature must specify target user, field problem, frequency, data owner, dependency, UX impact, security scope, acceptance tests and whether it is core or add-on.
- No “future” feature gets a placeholder nav item, schema or marketing promise merely because a competitor has it.
- Any request to remove/replace production data requires a separately approved backup, environment check and rollback plan. “Start from scratch” in this roadmap does not itself authorize deletion.

## 12. Official technical references

- React versions and current React documentation: https://react.dev/versions
- Vite guide and React TypeScript template support: https://vite.dev/guide/
- Tailwind CSS setup with Vite: https://tailwindcss.com/docs/installation/using-vite
- 21st.dev MCP usage, component search/install and Cursor support: https://docs.21st.dev/mcp
- Supabase Auth quickstart with React: https://supabase.com/docs/guides/auth/quickstarts/react
- Supabase Row Level Security: https://supabase.com/docs/guides/database/postgres/row-level-security
- Supabase database migrations: https://supabase.com/docs/guides/local-development/database-migrations
- Supabase Edge Functions: https://supabase.com/docs/guides/functions
- Supabase Storage access control: https://supabase.com/docs/guides/storage/security/access-control
- Nepal IRD CBMS API technical document for software developers: https://ird.gov.np/content/9052/cbmsapitechnicaldocumentfor/
- Nepal IRD electronic invoicing procedure (CBMS terminology/background): https://www.ird.gov.np/public/pdf/1491134985.pdf
- India IRIS IRP API integration guidance: https://einvoice6.gst.gov.in/content/api-integration/
- India IRIS IRP API integrator registration: https://einvoice6.gst.gov.in/content/kb/registration/
- India IRP onboarding APIs and taxpayer credential authorization: https://einvoice6.gst.gov.in/content/onboarding-apis-wiki/
- Expo Push Notifications overview (APNs/FCM intermediary): https://docs.expo.dev/push-notifications/overview/
- Expo Push service delivery/retry guidance: https://docs.expo.dev/push-notifications/sending-notifications/

These references support stack and integration planning; they do not certify FortSite’s security, accounting, engineering output or local regulatory compliance. Nepal IRD documentation found during planning includes a CBMS API technical-document page, and India IRIS

IRP documentation describes API-integrator and taxpayer-onboarding paths. These must be reconfirmed with the relevant authority/provider when Phase 67 begins; provider choices, eligibility, payloads, credentials and rules can change. Expo documents that push delivery passes through provider systems and can fail, so the durable Supabase inbox/outbox and retry handling remain mandatory.

**Final baseline:** Build FortSite from Phase 01 in dependency order. The Super Admin is the highest platform authority for company onboarding and subscriptions. Company roles remain separated and scoped. Construction operations, stock and accounting need reliable data boundaries. Supplier Marketplace is a separate optional add-on with messaging, managed RFQs, quote collection and mobile push notifications, but it never auto-creates a purchase order or payment. Automated takeoffs, rebar schedules and IFC clash detection are specialist opt-in modules with source traceability and qualified review. Country-specific live government billing connectors are the final Phase 67 after all other modules and general release tests have passed. Usability, security, engineering traceability and data integrity outrank feature count.